

DATA ANALYST PORTFOLIO PROJECT

Telco Customer Churn Analysis

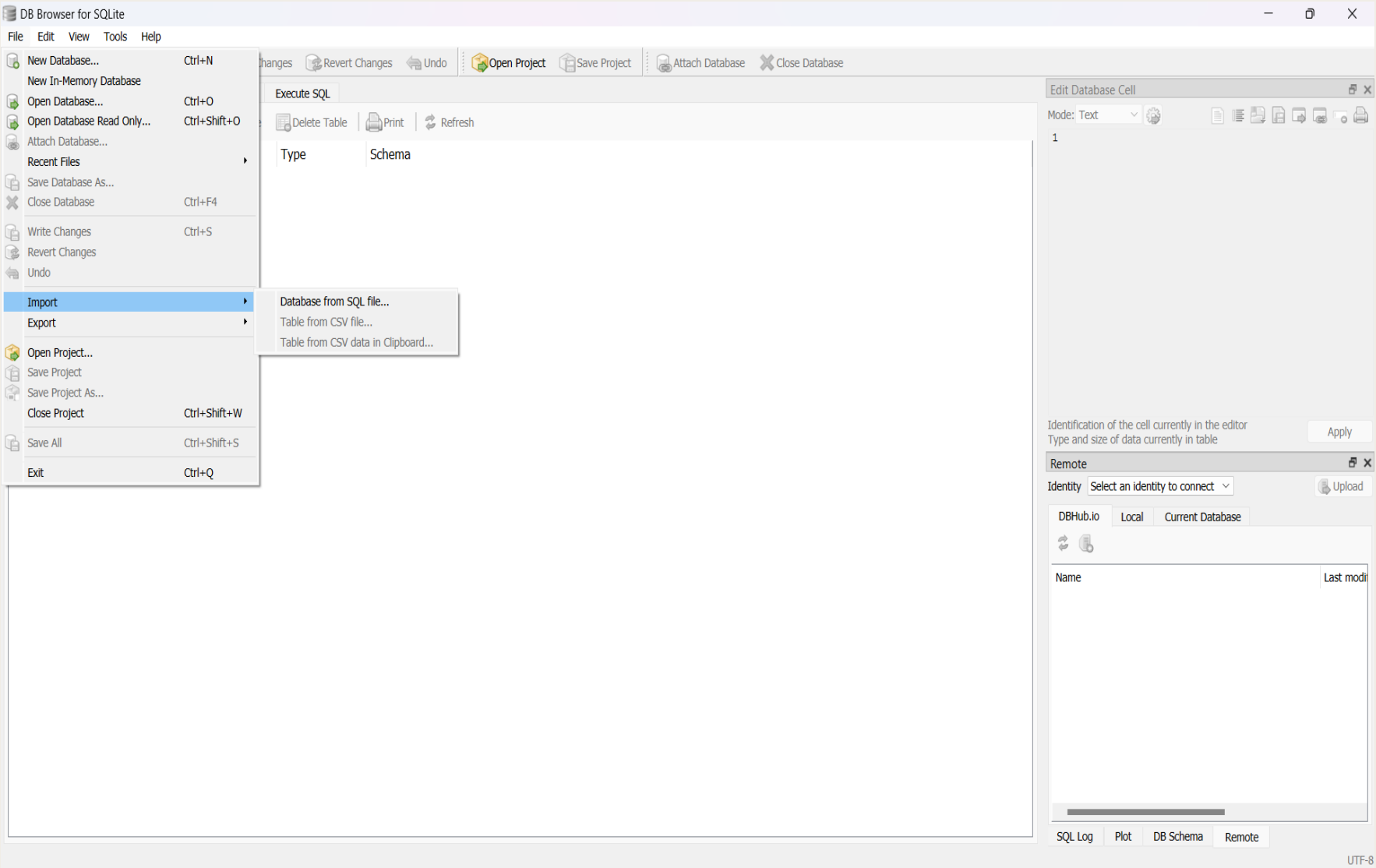
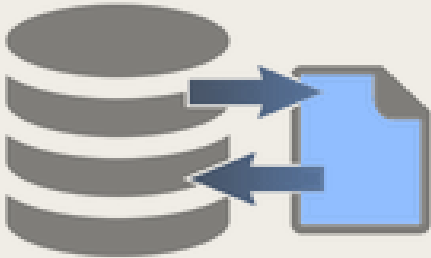
SQL + **Excel** Based Analysis

PRESENTED BY:

Pratiksha L Rohankar

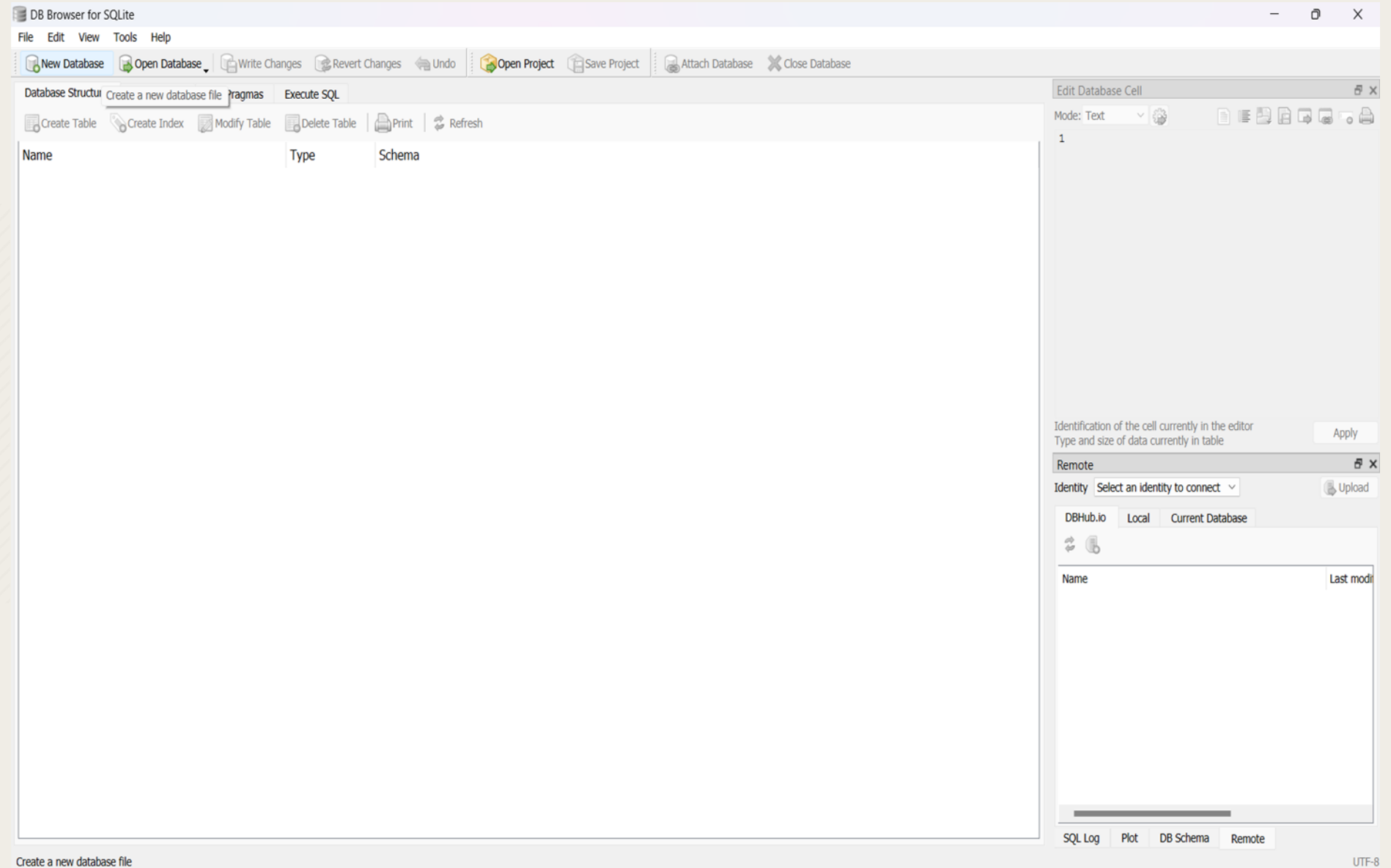
DB Browser for SQLite

IMPORT DATA



DB Browser for SQLite

CREATING DB



DB Browser for SQLite

WRITING QUERIE

DB Browser for SQLite - C:\Users\rohan\Downloads\My Data Analysis Projects\Telco Customer Churn Analysis Project\telco_churn.sqbpro [telco_customers.db]

File Edit View Tools Help

New Database Open Database Write Changes Revert Changes Undo Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

SQL 1

```
1 --All Customer Details
2 SELECT * FROM telco_customers;
3
4 --Total Customers
5 SELECT COUNT(*) FROM telco_customers;
6
7 --Unique Customers & Churned Customers
8 SELECT
9 COUNT(DISTINCT customerID),
10 SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS churned_customers
11 FROM telco_customers;
12
13 --Total Unique Customers
14 SELECT COUNT(DISTINCT customerID) FROM telco_customers;
15
16 --Churned vs Non-Churned Customers
17 SELECT Churn, COUNT(*)
18 FROM telco_customers
19 GROUP BY Churn;
20
21 --Total Customers
22 SELECT COUNT(*) AS total_customers FROM telco_customers;
23
24 --Churned Customers
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection
1	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	Yes	No
2	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	No	Yes
3	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	Yes	No
4	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	No	Yes
5	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	No	No
6	9305-CDSKC	Female	0	No	No	8	Yes	Yes	Fiber optic	No	No	Yes
7	1452-KIOVK	Male	0	No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No
8	6713-OKOMC	Female	0	No	No	10	No	No phone service	DSL	Yes	No	No

Edit Database Cell

Mode: Text

NULL

No cell active.
Type: NULL; Size: 0 bytes

Apply

Remote

Identity Select an identity to conn Upload

DBHub.io Local Current Database

Name

SQL Log Plot DB Schema Remote

UTF-8

CONNECTING Excel To DB Browser for SQLite



DATA PROCESSING



DATA VISUALIZATION

Telco Customer Churn Analytics Dashboard


Total Customers
7043


Churned Customers
1869


Churn Rate (%)
27%


Active Customers
5174

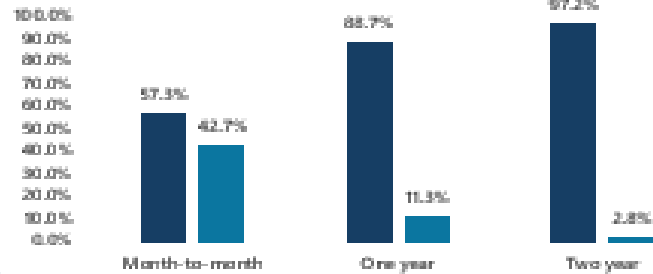

Avg Monthly Charges
₹ 64.76


Avg Tenure (Months)
32.3

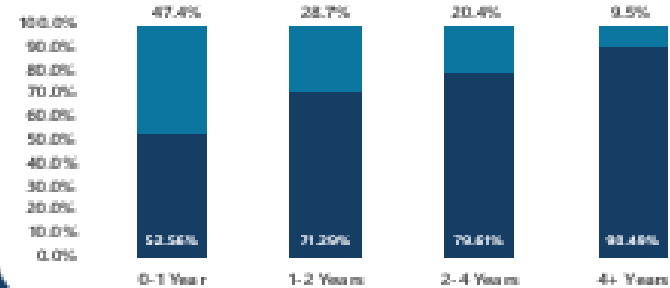

Total Customer Revenue
₹ 16.06M



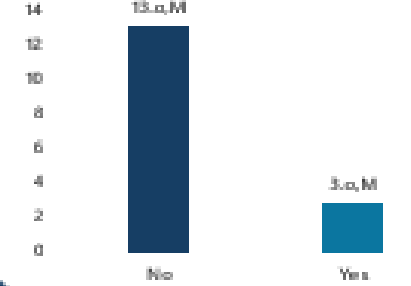
Churn by Contract Type



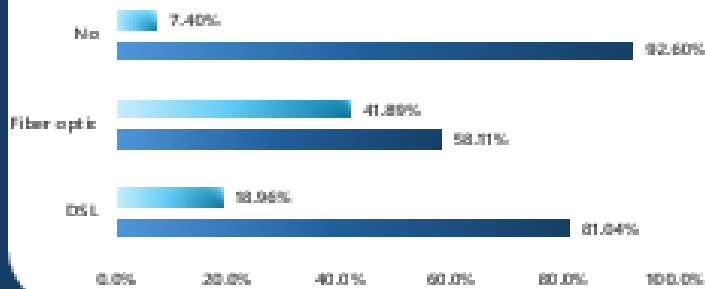
Churn by Tenure Band



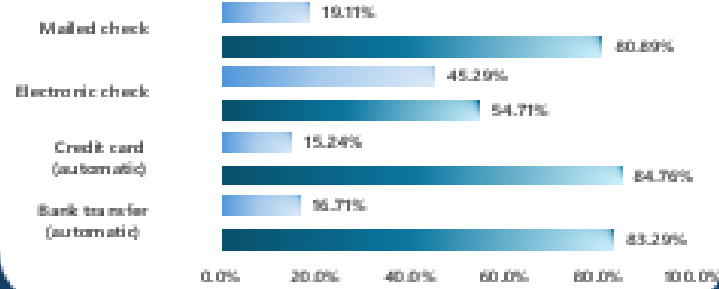
Revenue by customer Churn Status



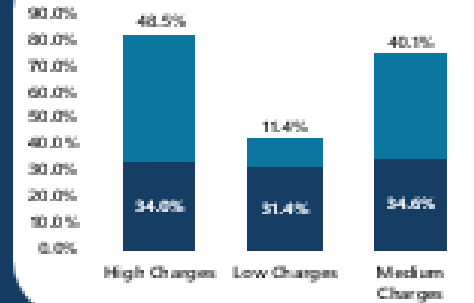
Churn by Internet Service



Churn by Payment Method



Customer Churn by Monthly Charge Level



Telco-Customers

churn_by_contract

churn_by_internet_service

churn_by_payment_method

churn_by_tenure_band

Churn_Analysis

PaymentMethod

No

Yes

PaymentMethod

Bank tra...

Credit c...

Electron...

Mailed ...

InternetService

DSL

Fiber optic

No

Contract

Month-to-month

One year

Two year

Executive Summary

Key Business Takeaways:

- Overall churn rate stands at **~27%**, indicating a **significant retention challenge**
- Churn is **not random** — it is strongly influenced by **pricing, contract type, and payment behavior**
- High-revenue customers are **more likely to churn**, increasing **revenue risk**
- Several churn drivers are **actionable**, enabling targeted retention strategies

Pricing Impact on Churn

Insight: Avg Monthly Charges – Churned vs Active Customers

- Churned customers have **higher average monthly charges** than active customers
- Indicates **price sensitivity**, especially among premium users
- High-paying customers may **not perceive sufficient value**

Business Implication:

- Offer **discounts for long-term contracts**
- Provide **loyalty rewards for long-tenure customers**
- Promote **auto-payment methods** to reduce churn
- Offer **discounts for high monthly charge customers**

Contract Type Drives Retention

Insight: Churn by Contract Type

- Month-to-month contracts show the **highest churn**
- Long-term contracts (1–2 years) show **significantly lower churn**
- Commitment length directly improves retention

Business Implication:

- Incentivize migration from **month-to-month** → **long-term contracts**
- Offer discounts or bundles during **renewal touchpoints**

Tenure-Based Churn Behavior

Insight: Churn by Tenure Group

- Churn is **highest among early-tenure customers**
- Customers with longer tenure show **higher loyalty and lower churn**
- Early experience is critical

Business Implication:

- Strengthen **onboarding & first 90-day experience**
- Trigger retention campaigns for customers in **0–12 month tenure**

Payment Method Risk

Insight: Churn Distribution by Payment Method

- Customers using **Electronic Check** exhibit the **highest churn**
- Auto-payment methods (credit card, bank transfer) show **lower churn**

Business Implication:

- Promote **auto-pay adoption**
- Provide incentives for switching away from high-risk payment methods

Revenue Loss Concentration

Insight: Revenue Loss Due to Churn by Payment Method

- A small number of payment methods account for **disproportionately high revenue loss**
- Revenue churn is **not evenly distributed**

Business Implication:

- Focus retention efforts where **revenue impact is highest**, not just churn count
- Revenue-weighted churn tracking should guide decisions

Monthly Charge Band Risk

Insight: Churn Share by Monthly Charge Band

- Customers in **higher monthly charge bands contribute more to churn**
- Indicates dissatisfaction among premium segments

Business Implication:

- Premium customers need **priority support & personalized offers**
- Flat retention strategies may fail for high-value customers

Internet Service Type

Insight: Churn by Internet Service Type

- **Fiber optic customers show higher churn** compared to DSL
- Higher service cost may raise expectations

Business Implication:

- Improve service reliability & support for fiber customers
- Reassess fiber pricing vs perceived value

THANK YOU